

RESEARCH ARTICLE

Extensive hybridization between two Andean warbler species with shallow divergence in mtDNA

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ABSTRACT

Studying processes acting on differentiated populations upon secondary contact, such as hybridization, is important to comprehensively understand how species are formed and maintained over time. However, avian speciation studies in the tropical Andes have largely focused on the role of topographic and ecological barriers promoting divergence in allopatry, seldom examining hybridization and introgression. We describe a hybrid zone involving 2 closely related Andean warblers (Parulidae), the Golden-fronted Redstart (*Myioborus ornatus*), and the Spectacled Redstart (*Myioborus melanocephalus*). Geographic ranges of these species abut near the Colombia-Ecuador border and many specimens from the region exhibit intermediate phenotypes, but a formal description of phenotypic variation in the contact zone was heretofore lacking. We collected specimens across a transect encompassing the area where ranges abut and areas where only “pure” parental phenotypes of *M. ornatus chrysops* and *M. melanocephalus ruficoronatus* occur. We described variation in plumage traits including patterns of head and ventral coloration and tail markings based on 321 specimens. To describe genetic variation in the contact zone and over a broader phylogeographic context, we used sequences of the mitochondrial ND2 gene for 219 individuals across the transect and the entire range of both species, including all subspecies, from Venezuela to Bolivia. We documented a hybrid zone ~200 km wide based on head coloration, where intermediate plumage phenotypes are most common and “pure” forms do not overlap geographically, consistent with extensive hybridization. Across the range of the *M. ornatus*–*M. melanocephalus* complex, mitochondrial genetic structure was shallow, with genetic breaks only coinciding clearly with one topographic feature. Such a low genetic structure is striking given the high diversity in plumage phenotypes and the current taxonomy of the group. Our phenotypic data suggest that barriers to hybridization are not strong, and allow us to postulate hypotheses to be tested using forthcoming genomic data.

Keywords: Andes, coloration, hybrid zone, mitochondrial DNA, *Myioborus*, Neotropics, phylogeography, speciation

LAY SUMMARY

- We studied plumage and genetic variation in 2 species of Andean warblers. We found low genetic differentiation despite high plumage diversity and an extensive area where the species interbreed.
- Studying processes occurring when formerly separate populations come into contact is useful to understand how species are formed. However, in the Andes, an area with high bird species diversity, few studies have focused on such processes.
- We found that populations of these warblers with marked differences in coloration do not have marked genetic differences, which may reflect the rapid evolution of color patterns in the group. We also found that our study species have interbred across ~200 km along the Andes, and that hybrids are common.
- Our work represents the first comprehensive sampling across the area where this species hybridize, and thus sets the base for future studies that might further our knowledge about the origin of species in Andean birds.

Amplia hibridación entre dos especies de parúlidos andinos con baja divergencia en ADN mitocondrial

RESUMEN

Estudiar procesos que actúan cuando poblaciones diferenciadas entran en contacto secundario, como la hibridación, es importante para tener una comprensión completa de cómo se originan las especies y cómo se mantienen en el tiempo. Sin embargo, muchos trabajos sobre especiación de aves andinas se han enfocado mayormente en entender el rol de barreras ecológicas y topográficas promoviendo divergencia en alopatria, pero pocos estudios han estudiado la hibridación e introgresión. En este estudio describimos una zona de hibridación de 2 especies de reinitas andinas, *Myioborus ornatus* y *Myioborus melanocephalus*. Estas especies se reemplazan geográficamente cerca de la frontera entre Colombia y Ecuador y existen múltiples especímenes con fenotipos intermedios, pero hasta ahora no existían descripciones de esta zona de contacto. Hicimos trabajo de campo para recolectar especímenes a lo largo de un transecto que incluía el área de reemplazo de las distribuciones de las dos especies y zonas donde sólo se encuentran individuos con fenotipos “puros”. Con base en 321 especímenes, describimos la variación en varios rasgos del plumaje, incluyendo la coloración de la cabeza y partes ventrales, y marcas de la cola. Para describir la variación genética en ADN mitocondrial en la zona de hibridación y en un contexto filogeográfico más amplio, usamos secuencias del gen mitocondrial ND2 para 219 individuos, incluyendo las regiones principales donde estas especies se encuentran desde Venezuela hasta Bolivia. Descubrimos que existe una zona de hibridación de aproximadamente 200 km de ancho según patrones de variación de coloración de la cabeza. Las formas intermedias son las más comunes y las formas “puras” no se traslapan geográficamente, lo que sugiere que la hibridación ha sido vasta. La estructura genética en el complejo *M. ornatus*–*M. melanocephalus* es bastante baja y coincide claramente sólo con un quiebre topográfico a lo largo de los Andes. La baja estructura geográfica es notable dada la alta diversidad en patrones de plumaje del grupo y la taxonomía actual. La variación fenotípica descrita sugiere que las barreras que impiden la hibridación son débiles y nos permite generar hipótesis que se pueden evaluar en estudios futuros usando datos genómicos.

Palabras clave: Andes, coloración del plumaje, zona de hibridación, ADN mitocondrial, Neotrópico, *Myioborus*, filogeografía, especiación

INTRODUCTION

When populations become geographically isolated, genetic and phenotypic differences tend to accumulate, potentially resulting in divergence in traits relevant for reproductive isolation (Coyne and Orr 2004, Price 2008). If formerly isolated populations come into contact and reproductive barriers are incomplete, then hybridization will occur (Harrison 1993), with different potential outcomes (Abbott et al. 2013). First, if isolation is incomplete and selection against hybrids is weak, then incipient reproductive barriers can break down leading to the collapse of differentiated populations into one (Taylor et al. 2006). Second, if selection against hybrids is strong, these reproductive barriers may strengthen via reinforcement (Dobzhansky 1937, Liou and Price 1994, Ortiz-Barrientos et al. 2009, Hudson and Price 2014). Finally, hybridization may persist over extended periods of time (Harrison 1986) if there is a balance between selection against hybrids and gene flow between parental populations (Slatkin 1973, Barton 1979, Barton and Hewitt 1985). Hybridization may also persist if selection favors hybrids in the area where they occur (Moore 1977, Wang et al. 1997, Good et al. 2000). Studying hybrid zones is therefore especially illuminating for understanding how species barriers are formed and maintained over time (Harrison 1993, Harrison and Larson 2016).

The tropical Andes are a biodiversity hotspot for multiple taxa (Brummitt and Lughadha 2003), including birds (Stotz et al. 1996, Hawkins et al. 2007, Fjeldså et al. 2012).

The rich Andean avifauna is characterized by pervasive intraspecific geographic variation in plumage (Remsen 1984, Graves 1988), with phenotypic and genetic variation often coinciding with physical barriers (Bonaccorso 2009, Chaves et al. 2011, Gutiérrez-Pinto et al. 2012, Valderrama et al. 2014, Benham et al. 2015, Winger and Bates 2015). Moreover, avian diversification in the Andes is thought to have resulted predominantly from allopatric speciation (García-Moreno and Fjeldså 2000, Weir 2009, Caro et al. 2013, Cadena and Céspedes 2020). Because Andean birds tend to have linear and elevationally restricted distributions, their ranges can be readily fragmented by gaps of unsuitable habitat (e.g., dry valleys for cloud forest species) or become disjunct because of local extinctions, leading to differentiation among isolated populations (Graves 1985, 1988). However, allopatric distributions caused by topographic or climatic barriers or by local extinctions are often not stable over time. For example, elevational movements of vegetation belts due to climatic fluctuations (Hooghiemstra and van der Hammen 2004, Ramírez-Barahona and Eguiarte 2013, Flantua et al. 2019) have likely promoted cycles of fragmentation and reconnection of suitable habitat for Andean species, potentially leading to a secondary contact between diverging populations (Vuilleumier 1969, Graves 1982). Although we have a growing understanding of how isolation across barriers contributes to phenotypic and genetic differentiation in the montane Neotropics, we know little about processes occurring when topographic and ecological barriers are overcome and closely related Andean lineages come into secondary contact (but see DuBay and

Witt 2014, Chattopadhyay et al. 2017, Morales-Rozo et al. 2017, Winger 2017).

The 12 species in the passerine genus *Myioborus* (Parulidae) inhabit montane areas across the Americas (Curson et al. 1994). Ten of these species occur at high elevations in the montane Neotropics, and most are distributed allopatrically in ranges separated by lowland gaps (Pérez-Emán 2005). Outstanding exceptions are *M. melanocephalus* (Spectacled Redstart) and *M. ornatus* (Golden-fronted Redstart), whose Andean distributions abut in southern Colombia and northern Ecuador (Curson 2010). Previous work revealed recent divergence (Lovette et al. 2010) and lack of reciprocal monophyly in mitochondrial DNA (mtDNA, cytochrome b, ND2, and ND3 genes) between these 2 species (Pérez-Emán 2005). The *M. ornatus*–*M. melanocephalus* complex, which radiated recently (<1.0 million years ago [mya]; Cuervo 2013), includes 7 subspecies with overall similarity in body plumage but marked geographic variation in head color patterns (Pérez-Emán 2005). Numerous records of individuals with intermediate plumage from the region where ranges abut (Zimmer 1949, Meyer de Schauensee 1964, Robbins et al. 1994, Cresswell et al. 1999, McCarthy 2006) suggest the existence of a hybrid zone between subspecies *M. ornatus chrysops* and *M. melanocephalus ruficoronatus*. However, detailed descriptions of patterns of variation in the area are lacking.

We collected specimens from multiple localities in Colombia and Ecuador to characterize the putative *Myioborus* hybrid zone using mitochondrial DNA and plumage coloration data. We supplemented our new samples with analyses of museum specimens that had been collected across the ranges of the hybridizing taxa since 1897. By examining plumage and mtDNA variation, we sought to answer the following questions: (1) what is the extent of hybridization between *M. ornatus chrysops* and *M. melanocephalus ruficoronatus*? (2) How do color and mtDNA vary across the hybrid zone? (3) What does the range-wide phylogeography of the *M. ornatus*–*M. melanocephalus* complex reveal about its evolutionary history?

METHODS

Study Species

Myioborus melanocephalus comprises 5 subspecies distributed along humid slopes of the Andes from southern Colombia to Bolivia (Curson et al. 1994). This species is characterized by a mostly black face and yellow spectacles, but subspecies differ in the amount of black on the submoustachial area and in the presence or absence of a rufous crown (Zimmer 1949, Curson et al. 1994). The southern subspecies *M. melanocephalus malaris*,

M. m. melanocephalus and *M. m. bolivianus* all lack the rufous crown, and replace each other geographically from south and east of the Marañón River Valley to central Bolivia (Figure 1A). Both northern subspecies have a rufous crown: *M. m. griseonuchus* of northern Peru (Piura and Cajamarca), and *M. m. ruficoronatus* of Ecuador and Colombia (Figure 1A). In turn, *M. ornatus* occurs in all 3 cordilleras of the Andes of Colombia and the Tamá massif west of the Táchira Depression in western Venezuela (Curson et al. 1994). This species is characterized by a yellow forehead and whitish ear patch, and comprises two subspecies: *M. o. ornatus*, from the Eastern cordillera and the Tamá massif, and *M. o. chrysops*, from the Western and Central cordilleras to extreme southern Colombia (Figure 1A). These taxa differ in their facial plumage pattern, with ear coverts and lores being yellow in *chrysops* and white in *ornatus* (Salvin 1837, Curson et al. 1994). All members of the complex occur in high-elevation habitats, such as cloud forests and scrub near the treeline, and occupy a similar elevation range (*M. ornatus*: 1,800–3,400 m, *M. melanocephalus*: 2,000–4,000 m; Curson 2010).

Collection of Specimens Across the Putative Hybrid Zone

We collected 113 specimens of *Myioborus* in 22 localities along a transect in Colombia and Ecuador, between July 2016 and March 2017, and in August 2018 (Figure 1B, Appendix Table 3). We selected field sites along this transect based on the location of previous reports of birds with intermediate plumage (Robbins et al. 1994, Cresswell et al. 1999, McCarthy 2006). Sites of collection along this transect were located roughly on the eastern side of the Andes, except for three sites in the central inter-Andean valley of Ecuador (Figure 1B; localities 2, 5, and 6). We also sampled putatively pure populations of *M. o. chrysops* in northern Antioquia and Caldas, Colombia, and of *M. m. ruficoronatus* in Tungurahua, central Ecuador (Figure 1B). Whereas the putatively pure populations of *M. o. chrysops* were sampled near the northern range limit of the species at the end of the Central Andes, the putatively pure population of *M. m. ruficoronatus* continues to range further south into southern Ecuador. Upon examination of specimens from southern Ecuador (e.g., from Morona Santiago province), the geographic limits between *M. m. ruficoronatus* and *M. m. griseonuchus* are not entirely clear, which makes it challenging to delineate the range of *M. m. ruficoronatus*.

We used specimens to obtain genetic and plumage coloration data. In most localities, we collected at least 4 specimens (Appendix Table 3; mean: 5.1; range: 1–8 specimens) except at La Sofía Road, which was excluded from cline construction because only one specimen was

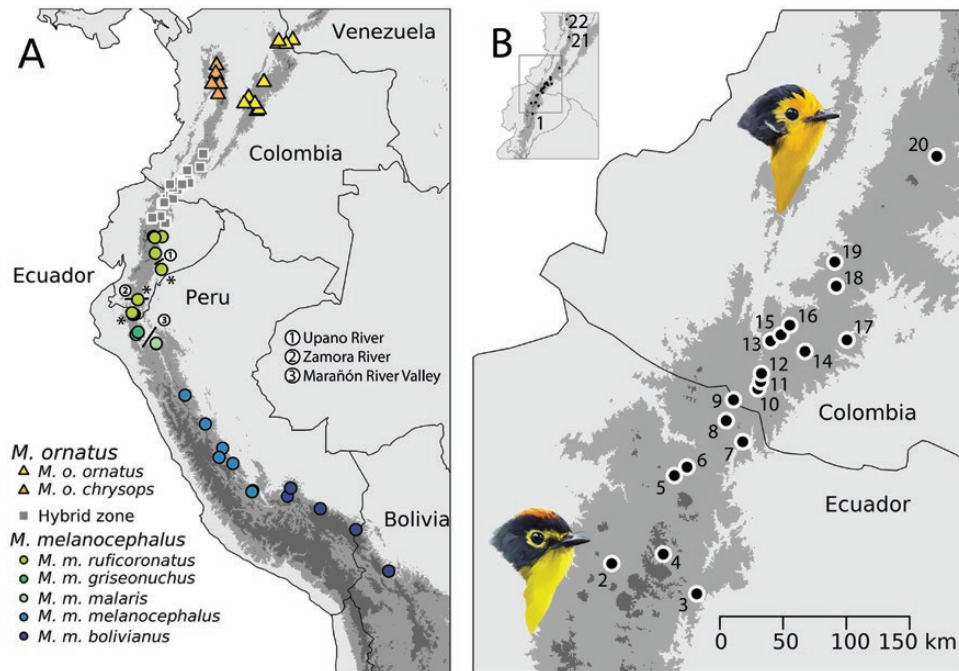


FIGURE 1. Study sites encompassing the distribution of *M. ornatus* and *M. melanocephalus*. In both maps, gray shades represent elevation, with medium gray corresponding to the elevation band inhabited by the study species (roughly 2,000–3,500 m). (A) Localities for which we obtained mtDNA sequences encompassing all currently recognized subspecies in the complex and hybrid zone birds, as defined for the genetic analyses. Asterisks (*) highlight localities in southern Ecuador where birds classified alternatively as *M. m. ruficoronatus* or *M. m. griseonuchus* for genetic analyses were collected. Numbers correspond to the approximate location of relevant topographical features. (B) Sampling localities across the putative hybrid zone for which we obtained mtDNA and color data. Localities captured by the inset were classified as hybrid zone birds in population genetic analyses. The numbers correspond to localities in Appendix Table 3. Illustrations by María Paula Bustamante.

collected (see below). Colombian specimens and tissue samples were deposited in the Museo de Historia Natural ANDES at Universidad de Los Andes (Bogotá, Colombia), and Ecuadorian specimens were exported and deposited at the museum of Instituto de Investigación de Recursos Biológicos Alexander von Humboldt (Villa de Leyva, Colombia; [Supplementary Material Table S1](#)). Duplicates of all tissue samples were deposited in the tissue collection of the Instituto Alexander von Humboldt. Vouchered sound recordings were also obtained for 47 (42%) of the specimens and deposited in the Macaulay Library ([Supplementary Material Table S1](#)).

Genetic Sampling (ND2 Mitochondrial Gene)

As a first approach to describe patterns of genetic variation across the geographic range of *M. ornatus* and *M. melanocephalus* and at their putative hybrid zone, we sequenced the NADH dehydrogenase subunit 2 (ND2) gene for specimens collected in this study and available from museum collections (total $n = 182$ for tissue samples, $n = 9$ for toepad samples; [Supplementary Material Table S2](#)). We extracted DNA from most tissue samples using a modified phenol-chloroform protocol ([Sambrook 1987](#)). For toepad samples, we extracted DNA using a QIAamp extraction

kit (Qiagen) in a lab where no other bird DNA is processed and under a UV-hood. We amplified ND2 from DNA extractions of preserved tissue samples following the protocol used by [Palacios et al. \(2019\)](#). To amplify the complete ND2 gene from toepad samples, we used 2 separate PCR reactions, both combining an external and internal primer (L5216 and H5766, L5758 and H1056U; [Sorenson et al. 1999](#), [Arbeláez-Cortés et al. 2014](#)). PCR mixes (25 μ L) for toepad samples consisted of 1.5 of DNA, 1.0 $\times$ of Taq buffer, 3.0 mM of $MgCl_2$, 0.2 mM of each dNTP, 0.48 μ M of each primer, 0.066 μ M of BSA and 1U of Taq DNA polymerase recombinant (Invitrogen). Then we used the following amplification conditions: 94°C for 5 min of initial denaturation; 35 cycles of 94°C for 45 s for denaturation, 51.9°C for 45 s for annealing, 72°C for 45 s for extension, and a final extension of 72°C for 7 min. We assembled and aligned consensus sequences using Geneious 9.0. ([Kearse et al. 2012](#)). We also compiled sequences from previous studies ($n = 28$; [Pérez-Emán 2005](#), [Cuervo 2013](#)). Our genetic sampling covered all taxa and main regions where the study species occur in Venezuela, Colombia, Ecuador, Peru, and Bolivia ([Figure 1](#), [Supplementary Material Table S2](#)).

Specimens were assigned to subspecies based on diagnostic plumage traits, and secondarily based on geographic

location (Salvin 1837, Zimmer 1949, Curson et al. 1994). A specimen was included in a “hybrid zone” group irrespective of its plumage phenotype if it was collected near a locality where intermediates might occur (Figure 1). This area was defined upon examination of the series of specimens we obtained and others collected prior to our study. Consequently, for genetic structure analyses, individuals from the hybrid zone correspond to a geographic group and not necessarily to a group of individuals with intermediate plumage phenotype. We found discordance between the reported geographic limits for *M. m. griseonuchus* and *M. m. ruficoronatus* (Zimmer 1949) and the distribution of observed plumage variation in specimens we directly inspected from across southern Ecuador. Specifically, we found that *M. m. griseonuchus*, as a phenotypic group, might reach outlying ridges in Morona Santiago where previous work suggested one should find *M. m. ruficoronatus*. Given this discordance between the literature and our own assessments, we ran some genetic analyses in 2 ways; one grouping specimens from southern Ecuador (Figure 1A, Supplementary Material Table S2) as part of *M. m. griseonuchus* (guided by our own assessment), and the other considering these specimens as *M. m. ruficoronatus* (guided by historical reports, Supplementary Material Table S2). Sample sizes per group following the latter classification were: *M. melanocephalus bolivianus* ($n = 12$), *M. m. melanocephalus* ($n = 10$), *M. m. malaris* ($n = 8$), *M. m. griseonuchus* ($n = 12$), *M. m. ruficoronatus* ($n = 18$), *M. o. ornatus* ($n = 27$), *M. o. chrysops* ($n = 15$), and hybrid zone birds ($n = 117$). We were able to obtain high-quality sequences of at least 1,026 base pairs (bp) for most samples; 22 sequences with <1,026 bp of length were excluded from some analyses (see below).

Phylogeography and genetic structure in the *M. ornatus*–*M. melanocephalus* complex. To examine relationships among populations across the distribution of *M. ornatus* and *M. melanocephalus*, we inferred ND2 gene trees with maximum-likelihood and Bayesian methods using unique haplotypes identified using DnaSP 6 (Librado and Rozas 2009) and *Myioborus albifrons*, the sister species to the *M. ornatus*–*M. melanocephalus* complex, as outgroup (Perez-Eman 2005). All sequences ($n = 219$; Supplementary Material Table S2) were incorporated in this analysis, including those with missing data (28–256 bp). Prior to conducting phylogenetic analyses, we identified unique haplotypes using DnaSP, and then obtained an alignment with a single sequence representing each unique haplotype (102 including the outgroup, and considering positions with missing data). The maximum-likelihood analysis was conducted using RaxML 8. (Stamatakis 2014) with a GTR+ Γ model of substitution with 25 gamma rate categories and 1,000 bootstrap replicates. The Bayesian analysis was done using MrBayes 3.2.2 (Ronquist and

Huelsenbeck 2003) with a GTR+ Γ , HKY and GTR+ Γ models of substitution, respectively, for each codon as suggested by PartitionFinder. We ran four MCMC chains of 50 million generations sampling every 1,000 generations and discarding the first 50% as burn-in. For both analyses, we implemented a partition scheme by codon as suggested by PartitionFinder 2 (Lanfear et al. 2017). We assessed convergence by examining trace plots and effective sample sizes in Tracer 1.6 (Rambaut et al. 2014). All analyses were run through the CIPRES platform (Miller et al. 2010).

To further describe relationships among ND2 haplotypes, we built a median-joining network (Bandelt et al. 1999) in PopArt 1.7. (Leigh and Bryant 2015). We built the network defining $\epsilon = 0$. This analysis and subsequent population genetic analyses were based on an alignment of 197 sequences with 1026 unambiguous positions. To test for population structure among subspecies and hybrid zone birds we calculated F_{ST} using functions implemented in the *adegenet* 2.1.1 for R (Jombart 2008) assessing significance with 2,000 permutations. We also calculated Nei genetic distances between these groups in *adegenet* 2.1.1 for R (Jombart 2008). Lastly, we estimated nucleotide diversity and Tajima's D statistics (Tajima 1989) per group, using functions implemented in the R package *pegas* 0.13. (Paradis 2010). To perform genetic analyses we also used various functions implemented in the package *ape* 5.3. (Paradis and Schliep 2019).

Plumage Coloration

We described variation in plumage across the hybrid zone by analyzing multiple traits that vary within the *M. ornatus*–*M. melanocephalus* complex (Fjeldsø and Krabbe 1990, Curson et al. 1994). This included the extent of the rufous patch on crown, extent of black in forecrown and on sides of the face, extent of whitish ear patch, extent of white on tail feathers, and the hue of underparts. We included juveniles with adult plumage in analyses (determined by the presence of Bursa of Fabricius and incomplete skull pneumatization, Winker 2000); excluding them did not seem to affect results in preliminary analyses. We did not run analyses separately for males and females because of the lack of noticeable sexual dichromatism, and because sample sizes for females were small.

Head color pattern. To assess variation in patterns of head coloration, we established categories based on the presence and extent of color plumage patches (Table 1). We then standardized these scores and obtained a per-individual plumage hybrid index by averaging the standardized scores per patch (Rohwer and Wood 1998, Shriver et al. 2005, Toews et al. 2011), with 1 representing a typical *M. m. ruficoronatus* and 0 representing a typical *M. o. chrysops*. Scoring categories (Table 1) were

established by a preliminary inspection of the series collected in this study and specimens in other museums (ICN, FMNH, IAvH, MUN; see acronyms in [Supplementary Material Table S3](#)). Two observers independently scored most specimens included in cline analyses (see below; 107 out of 120) and we used the average for analyses. The remaining specimens were scored by one of the observers. Although not identical, scores were highly correlated between observers (Pearson's $r = 0.99$). We also obtained head-color scores and body-color data (see below) from additional specimens collected between 2006 and 2015 (available at the IAvH and MUN collections, see [Supplementary Material Table S1](#) for acronyms), which allowed us to increase sample sizes for two localities ($n = 11$; Alto de Daza and Laguna de la Cocha, [Supplementary Material Table S1](#)).

In addition to scoring specimens collected by us and included in the geographic cline analyses (see below), we examined specimens collected from 1897 to 2015 ($n = 184$, [Supplementary Material Table S3](#)). This procedure allowed us to qualitatively describe the distribution of hybrids while considering localities not included in our field sampling, and to potentially detect plumage phenotypes not observed in the specimen series we collected and included in the cline analyses. Examined specimens are in the following museums: FMNH, ICN, IAvH, AMNH, MUN, and ANSP (see [Supplementary Material Table S3](#) for acronyms). One observer familiar with the scoring system obtained hybrid indices for all specimens for which all head patches were assessed with certainty ($n = 170$).

Additionally, we aimed to assign all specimens to either taxon or hybrids, when mixed facial traits were present. We classified as “uncertain” those specimens with no clear evidence of mixed facial traits, but which we could not confidently assign to either taxon. We also used this classification for specimens inspected through photographs, deposited at MLZ and USNM ($n = 16$; [Supplementary Material Table S3](#)), for which we did not assign a hybrid index given our inability to examine auricular feathers in detail. We also employed this categorization for specimens included in the cline analyses, as a complementary way to describe the geographic distribution of individuals with intermediate plumage.

Underparts coloration. Coloration of the underparts has been described as bright yellow in *M. m. ruficoronatus* (Fjeldså and Krabbe 1990), and as warm yellow (Fjeldså and Krabbe 1990) or with an orange tinge (Curson 1994) in *M. o. chrysops*. To quantify variation in color hue in the throat and chest we used reflectance spectrometry. Reflectance measurements were taken from specimens using an Ocean Optics USB4000 spectrophotometer and a DH-2000 deuterium halogen light source coupled with an optic fiber

TABLE 1. Scoring categories established to calculate the hybrid index. Higher scores correspond to typical *M. melanocephalus ruficoronatus* traits and lower to *M. ornatus chrysops*.

Score	Description
Rufous crown ^a	
0	No visible rufous crown. Hint of rufous on the base of black feathers might be present.
1	Few visible rufous feathers.
2	Small crown, almost like a band (3–5 mm).
3	Full rufous crown (>6 mm).
Black in forehead ^a	
0	No black on the front of the rufous crown. Rufous is in direct contact with a yellow forehead.
1	Few spots of black between the rufous and yellow forehead. Extensive yellow on the forehead.
2	The narrow black band in front of the rufous crown, or transition between rufous and yellow heavily spotted with black.
3	The intermediate black band in front of the rufous crown. A continuous band of yellow feathers that separates black from the culmen is present.
4	The wide black band in front of the rufous crown. Black is in direct contact with culmen, but a few dispersed yellow feathers might be present.
Black stripes under eye	
0	Sides of the face completely yellow.
1	Barely visible black stripe under eye. Often incomplete, being only present near the beak.
2	Clear black stripe under the eye, always continuous from the beak but narrow.
3	Wide and complete black stripe (>1.5 mm) under eye and wider near the beak.
Whitish ear patch	
0	White auricular composed by >10 whitish feathers inserted next to tympanum forming a continuous patch.
1	White auricular composed by 3–6 whitish feathers inserted next to tympanum.
2	1–2 whitish feathers inserted next to tympanum.
3	No whitish feathers inserted next to the tympanum.

^aBecause these traits were nonindependent (absence of a rufous crown implied the absence of a black band in front of it), we calculated the hybrid index by treating them as one trait. Specifically, we summed the score for both patches and then divided by 7 (maximum value) to obtain a “crown” score.

QP400-2-UV-VIS with a 400 μm diameter. The spectrophotometer was calibrated using a white standard. Each color patch was measured three times per individual and then the reflectance spectra were averaged using functions implemented in *pavo* 2.4.0. for R (Maia et al. 2013). Prior to averaging, we processed each reflectance spectra using the *prospec* function, converting all negative values of reflectance to zero and smoothing the curves using a smoothing parameter of 0.2.

We calculated 6 variables to describe reflectance spectra without assuming specific visual models (Endler 1990, Montgomerie 2006). First, we calculated values of chroma for the spectral range corresponding to yellow (S1.yellow; 550–625 nm), red (S1.red; 605–700 nm), and UV (S1.UV; 300–400 nm) (Montgomerie 2006, Maia et al. 2013; Appendix Figure 7). We selected these ranges because they capture the most prominent peaks in the reflectance spectra we analyzed (see examples in Appendix Figure 7). Chroma is calculated as a proportion of the reflectance under a given wavelength range relative to the total reflectance of the curve. We also calculated hue, brightness, and chroma using a segment classification approach (Endler 1990; Appendix Figure 7). Some specimens were excluded from this measurement due to heavy molt (for throat $n = 81$, for chest $n = 100$, Supplementary Material Table S1).

Tail color pattern. According to previous descriptions, there is substantial variation in the amount of white in the tail within the *M. ornatus*–*M. melanocephalus* complex (Fjeldså and Krabbe 1990, Curson et al. 1994). To describe variation in tail coloration we measured the extent of white on both vanes of the three outer rectrices with a dial caliper with a precision of 0.1 mm and obtained an average value per feather (Pérez-Emán et al. 2010). We then summed the 3 average values per feather and divided by the sum of the total vane length of these 3 feathers. We excluded several individuals in active molt or worn plumages at the time of collection (total included $n = 79$, Supplementary Material Table S1).

Geographic Cline Analyses and Distribution of Phenotypes

We performed geographic cline analyses to examine the extent of hybridization between *M. ornatus* and *M. melanocephalus*, and to describe how plumage traits vary across their hybrid zone. To assign distances along the transect, we fitted a linear regression between latitude and longitude for each of our sampling localities (Appendix Figure 8). We used the fitted line as a reference transect of geographic distance by connecting to it every locality with a perpendicular line, which allowed us to measure the location (distance in kilometers [km]) of every locality along this reference transect with respect to the southernmost locality (Parroquia El Triunfo, Tungurahua, Ecuador; Appendix Table 3). Only specimens collected in 2016–2018 for this study and in the same localities in previous years (2006 and 2015) were included in cline analyses (Supplementary Material Table S1).

We fitted traits (i.e. hybrid index for head coloration) showing differences between the extremes of the transect to various equilibrium cline models (Szymura and Barton 1986, Gay et al. 2008) implemented in the R package *HZAR* 0.2.5. (Derryberry et al. 2014). Specifically, we fitted 5

models that estimate cline width (w) and center (c), and have fixed values for mean and variance of the trait values at the cline extremes (μ_L , μ_R , var_L , var_R). Models varied in the parameters describing the exponential decay curves that were estimated (δ and τ). We fitted the following models: (1) a model that only estimates center and width of the cline (Model I; c , w and $varH$); (2) a model that estimated decay curve parameters for the right tail (Model II; c , w , $varH$, δR and τR); (3) a model that estimated decay curve parameters for the left tail (Model III; c , w , $varH$, δL and τL); (4) a model that estimated decay curve parameters for mirrored tails (Model IV; c , w , $varH$, δM and τM); and (5) a model that estimated decay curves parameters independently for each tail (Model V; c , w , $varH$, δR , τR , δL and τL). Runs for all models were done with a tune value of 1.5, except for model V, for which the tune was adjusted to 1.2. We ran 3 chains of 1 million iterations per model discarding the first 10% as burn-in, and assessed convergence using trace plots. All runs were performed randomizing starting parameters. We chose the best-fit cline model for each trait based on AICc values.

We also described the frequency distribution of head plumage phenotypes (i.e. the trait that more clearly distinguishes hybridizing taxa) because it may provide information on the type of selection acting on hybrid zones (Gay et al. 2008). A bimodal distribution where both pure forms are common in the center is expected under disruptive selection and assortative mating, whereas a unimodal distribution in the center is expected if intermediates are the most common as a result of extensive hybridization (Gay et al. 2008). We described the distribution of the hybrid index by only considering localities within the estimated width around the center estimated with *HZAR*, and evaluated whether it followed a unimodal distribution using a Hartigan's Test for Unimodality as implemented in package *diptest* 0.75.7. (Maechler 2016).

As an alternative way to describe geographic distribution in head coloration, we used information from specimens collected by us and others already deposited in collections. First, we assigned geographic coordinates to all historical specimens. To associate geographic coordinates to these specimens we used the locality description, excluding all specimens with localities corresponding to provinces/departments or countries, or with dubious locations (e.g., towns in the lowlands, far from elevations where these birds occur). Historical collecting localities were georeferenced using gazetteers (Paynter and Traylor 1977, Paynter et al. 1997) and other sources (Supplementary Material Table S3). Then, using hybrid indices and associated geographic coordinates, we obtained a raster describing the spatial variation of phenotypes. Specifically, we created a grid of $0.25^\circ \times 0.25^\circ$ and averaged the hybrid index values of specimens present in each pixel, using functions implemented in

the package *raster* 3.0.7. (Hijmans and van Etten 2012). Because of the lack of a tight systematic sampling across the entire hybrid zone prior to this study, we did not aim to compare historic and contemporary patterns of variation.

RESULTS

Phylogeography and Genetic Structure

Genetic structure in the *M. ornatus*–*M. melanocephalus* complex along the Andes was shallow, and seemingly related to only one topographic and ecological barrier. We recovered two well-supported groups in the Bayesian gene tree, but with low bootstrap support in the maximum-likelihood tree (Figure 2A): (1) a southern clade including all but 3 of the specimens from south of the Marañón River Valley located in northern Peru (corresponding to subspecies *M. m. malaris*, *M. m. melanocephalus* and *M. m. bolivianus*), and (2) a clade with high nodal support including all *M. m. griseonuchus*, some specimens from southern Ecuador (i.e. from Loja, Zamora-Chinchipe, Morona Santiago), a few hybrid zone specimens, and 3 *M. m. malaris* specimens that did not cluster with the southern clade (Figure 2A, haplotypes 50, 51, and 52). The remaining specimens, including *M. o. ornatus*, *M. o. chrysops*, *M. m. ruficoronatus*, and most hybrid zone birds, did not form clades associated with geography in either the Bayesian or maximum-likelihood trees.

The haplotype network showed 2 genetic clusters consistent with well-supported groups in the Bayesian gene tree (Figure 2A, groups 1 and 2), and a third cluster including the northern specimens corresponding to *M. o. ornatus*, *M. o. chrysops*, most *M. m. ruficoronatus* as well as most hybrid zone birds (Figure 2B, hereafter *M. ornatus* + *M. m. ruficoronatus* group), which also shared the most common haplotype. The remaining hybrid zone specimens were grouped with *M. m. griseonuchus* and some specimens of *M. m. ruficoronatus*, as in the Bayesian tree (Figure 2A, group 2). These three clusters, which are broadly congruent with geography, are separated from each other only by at most 4 mutational steps (Figure 2B) and exhibit high haplotypic diversity. We observed other smaller clusters with a comparable level of differentiation yet not having a clear association with plumage and geographic groups. For example, one cluster grouped specimens from the hybrid zone from southern Ecuador (haplotypes 33, 34, and 35 which are marked by an arrow, Figure 2B).

Both F_{ST} and Nei genetic distances indicated low genetic differentiation within the *M. ornatus* + *M. m. ruficoronatus* group (Table 2, Appendix Table 4). F_{ST} values were lower than 0.1 for all pairwise comparisons (Table 2, Appendix Table 4). Individuals from the hybrid zone, as a geographic group, were not differentiated from *M. m. ruficoronatus* nor from either subspecies

of *M. ornatus* (Table 2, Appendix Table 4). Conversely, we found higher values of F_{ST} for most of the pairwise comparisons involving the southern subspecies (*M. m. griseonuchus*, *M. m. malaris*, *M. m. melanocephalus*, *M. m. bolivianus*; Table 2, Appendix Table 4). Sample sizes for these southern groups were small, though, especially for *M. m. malaris*. Because we found no haplotypes with high frequency in one hybridizing taxon and low in the other, we were unable to build a haplotype frequency cline across the sampling transect. The frequency of the most common haplotype, which was shared by *M. o. chrysops*, *M. m. ruficoronatus* and hybrid zone birds, did not vary clinally across the hybrid zone.

Nucleotide diversity was relatively low for all groups (i.e. subspecies and hybrid zone specimens), ranging from 0.02 to 0.07 (Appendix Table 5), and being lowest in both subspecies of *M. ornatus* and the hybrid zone geographic group and highest in *M. m. malaris*. In contrast, haplotype diversity was relatively high in all groups, ranging from 0.75 to 1 and being highest in *M. m. malaris*. Tajima's D was negative for all groups under both classification schemes, although most values were non significantly different from zero, with the exception of the hybrid zone group (Appendix Table 5). However, it should be noted that sample sizes per group, especially the southern subspecies, are relatively low.

Plumage Coloration Across the Hybrid Zone

We found a wide variety of plumage phenotypes including different combinations of color traits typical of *M. o. chrysops* and *M. m. ruficoronatus* (Figure 3). For example, some individuals had an extensive rufous crown as in *M. m. ruficoronatus* and a yellow forehead typical of *M. o. chrysops*, whereas other specimens also had a hint of black in the forecrown and on the sides of the face (Figure 3A). The amount of black on the forehead and sides of the face, when present, varied substantially among individuals (Figure 3A and 3C). Individuals of *M. m. ruficoronatus* away from the hybrid zone still showed variation in the amount of black on the sides the face, even among specimens from nearby localities. Some individuals showed either a complete or partial rufous crown like in *M. m. ruficoronatus* but also showed a clear withish ear patch, like in *M. o. chrysops*. Also, a few specimens collected near the northern reaches of the hybrid zone had a typical *M. o. chrysops* forehead and crown but lacked completely the withish auricular characteristic of this taxon. Although variation in the rufous crown did not appear to be continuous, 2 intermediate states were found in between the extremes of being fully present or absent: a few rufous feathers visible or a fully formed but much smaller crown than in typical *M. m. ruficoronatus* (Figure 3B). Several specimens of *M. o. chrysops* away

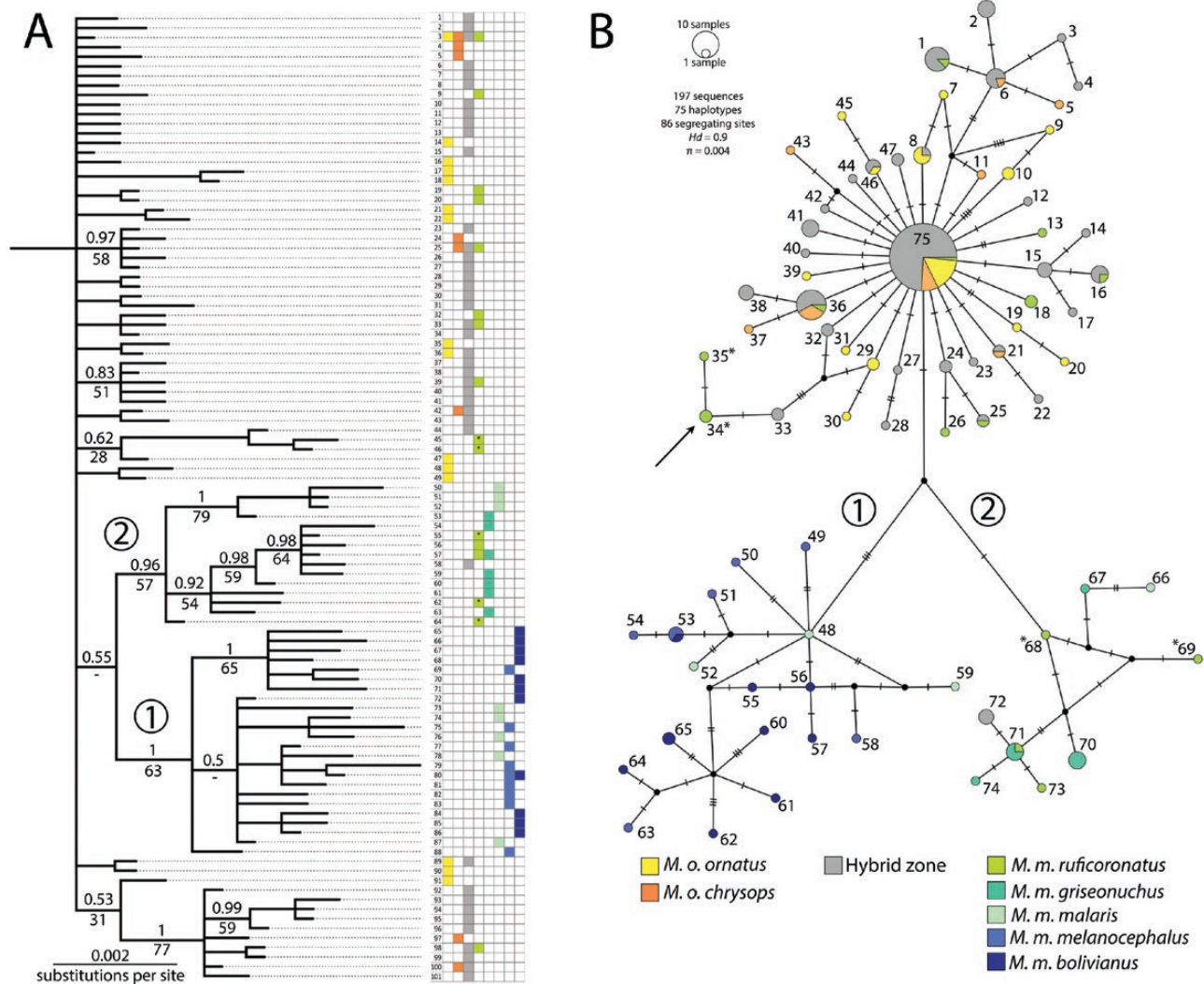


FIGURE 2. Genetic structure in the *M. ornatus*–*M. melanocephalus* complex inferred from ND2 sequences. **(A)** Bayesian 50% majority-rule tree for unique haplotypes, showing high posterior probability values for 2 clades: one including all specimens from *M. m. boliviensis*, *M. m. melanocephalus*, and most of *M. m. malaris* (1) and the other including all *M. m. griseonuchus*, *M. m. ruficoronatus*, three *M. m. malaris*, and some hybrid zone birds (2). Posterior probability showed above lines and bootstrap values for the maximum likelihood, below. Tip label corresponds to unique haplotype numbering that is associated with specimens in the [Supplementary Material Table S2](#). The outgroup (*M. albifrons*) is not shown for clarity. **(B)** MJ haplotype network revealed 3 clusters that are broadly congruent with geography and with the gene tree. Mutational steps are shown as hatches along lines and inferred haplotypes as filled black circles. Numbers correspond to haplotype number in the [Supplementary Material Table S2](#) and do not correspond to numbering in panel (A). Asterisks (*) in both panels highlight birds from southern Ecuador that were classified as either *M. m. ruficoronatus* or *M. m. griseonuchus* in some analyses.

from the hybrid zone, and even of *M. o. ornatus*, had black feathers in the crown with a rufous base, but this was concealed and not evident without a detailed examination. We also found a clear intermediate state for the white ear patch, with some specimens having 1–2 whitish feathers per side, whereas typical *M. o. chrysops* had more than 10 and typical *M. m. ruficoronatus* lacked feathers with such coloration (Figure 3C).

Individuals with intermediate plumage were common along the sampling transect. For example, in localities 13–16 all individuals were clearly intermediate (hybrid index = 0.27–0.79; Figure 1B). Pure plumage forms (*M. o. chrysops* and *M. m. ruficoronatus*) were never observed or collected together in the same locality. Head coloration as measured by the hybrid index showed a clear pattern of sigmoidal variation across the transect (Figure 4).

TABLE 2. Genetic differentiation between groups in the *M. ornatus*–*M. melanocephalus* complex. Numbers below the diagonal correspond to the F_{ST} values that are in bold if P value < 0.05. Above the diagonal, mean Nei-genetic distances between defined groups. These genetic distances and F_{ST} values were calculated using the classification for *M. m. griseonuchus* and *M. m. ruficoronatus* based on historical reports.

	<i>M. o.</i> <i>ornatus</i>	<i>M. o.</i> <i>chrysops</i>	Hybrid zone	<i>M. m.</i> <i>ruficoronatus</i>	<i>M. m.</i> <i>griseonuchus</i>	<i>M. m.</i> <i>malaris</i>	<i>M. m.</i> <i>melanocephalus</i>	<i>M. m.</i> <i>bolivianus</i>
<i>M. o. ornatus</i> ($n = 26$)	–	0.006	0.004	0.012	0.075	0.071	0.095	0.095
<i>M. o. chrysops</i> ($n = 15$)	0.068	–	0.002	0.012	0.076	0.072	0.096	0.097
Hybrid zone ($n = 109$)	0.026	0.009	–	0.009	0.070	0.069	0.094	0.095
<i>M. m. ruficoronatus</i> ($n = 16$)	0.083	0.087	0.035	–	0.047	0.065	0.094	0.093
<i>M. m. griseonuchus</i> ($n = 9$)	0.407	0.507	0.171	0.202	–	0.087	0.125	0.124
<i>M. m. malaris</i> ($n = 4$)	0.256	0.365	0.085	0.161	0.386	–	0.016	0.034
<i>M. m. melanocephalus</i> ($n = 8$)	0.418	0.504	0.191	0.297	0.523	0.086	–	0.026
<i>M. m. bolivianus</i> ($n = 10$)	0.424	0.489	0.220	0.305	0.495	0.149	0.152	–

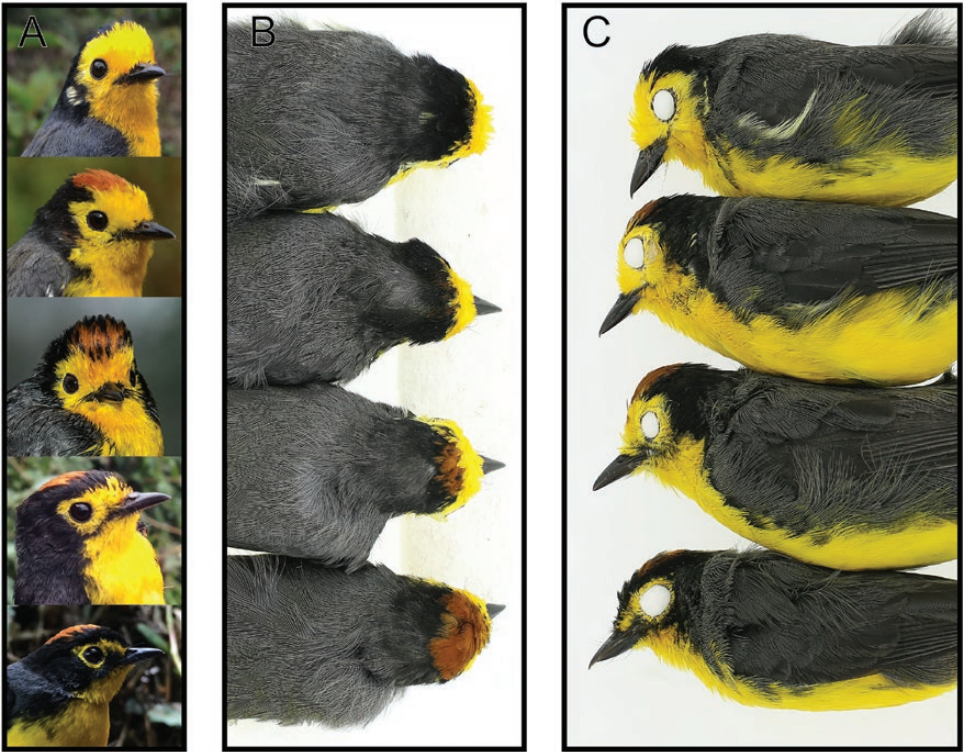


FIGURE 3. A sample of the diversity of plumage phenotypes presents across the hybrid zone. (A) Head color patterns, from top to bottom: typical phenotype of *M. o. chrysops*; intermediate phenotype, showing a solid yellow face and forehead and an extensive rufous crown; intermediate with only a hint of black in the rear of the yellow forehead; intermediate with a moderate amount of black between surrounding the rufous crown, and typical *M. m. ruficoronatus*. (B) Variation in crown coloration, ranging from non-visible rufous feathers (top) to a full rufous crown (bottom). (C) Variation in the presence and extent of the withish ear patch, and width of the black stripe under the eye, ranging from specimens showing a withish ear patch and no black stripe under the eye to specimens with no withish ear patch and a solid black stripe under the eye. Photo credits: Paulo Pulgarín (A), David Ocampo (A), and Laura Céspedes (A, B, C).

According to variation described by the hybrid index, the hybrid zone is 217 km wide (95% confidence interval [CI]: 192–247 km) and is centered at 307 km (95% CI: 299–316 km) along the sampling transect, around localities 14–16

(Figure 1B). Width and center values were estimated based on the simplest model which had the lowest AIC_c value (model I = –233.50, model II = –231.31, model III = –229.18, model IV = –229.30, model V = –226.62).

The distribution of the hybrid index in the hybrid zone (localities 8–19) was clearly unimodal: individuals with intermediate phenotypes were most common (Figure 4B, Hartigan's dip test P -value = 0.143).

Considering plumage variation of historic and modern specimens from across the hybrid zone, individuals with intermediate phenotypes are restricted to extreme southern Colombia and northern Ecuador, and occur on both slopes of the Andes (Figure 5). We found hybrid historical specimens in the western ridges of the Colombia-Ecuador border (Figure 5). However, hybrid indices in that region appear to be higher than in the eastern slope at the same latitude within our sampling transect. Moreover, of the 26 specimens examined in the west, very close to the Colombia-Ecuador border, only 3 were confidently categorized as intermediates, and most were categorized as “pure” *M. m. ruficoronatus* ($n = 17$, Figure 5B in the area encircled by ellipse, Supplementary Material Table S3). In contrast, localities on the eastern slope of the Andes, where we concentrated our fieldwork, harbor a high proportion of intermediates. Historical specimens also revealed the southernmost location for clearly intermediate specimens (2 specimens in Oyacachi, Pichincha; hybrid index = 0.70–0.86, arrow in Figure 5B).

All colorimetric variables describing underpart coloration are strongly correlated to each other (Appendix Table 6), with most of them showing differences between the extremes of the sampling transect (Appendix Figure 9). After exploring variation in each variable we focused on hue because it seemingly followed a clinal trend, at least for the chest (Figure 6, Appendix Figure 9). The estimated cline width for chest hue was 226 km (95% CI: 161–333 km) with a center around 324 km (95% CI: 287–381 km, Figure 6B). Confidence intervals around cline width and center of chest hue overlap entirely with those of the head coloration cline. For the throat patch, the center (–14 to 347 km) and width (237–1030 km) confidence intervals are remarkably wide despite cline analyses seemingly achieving convergence. The estimates around the width of the throat cline include values larger than the transect length itself, potentially reflecting that a sigmoidal function does not fit the data well and variation might be more linear. For both chest and throat, the model with the lowest AIC_c was the simplest model (model I = 532.49, model II = 536.87, model III = 536.69, model IV = 536.68, model V = 541.33 for chest; model I = 465.10, model II = 467.96, model III = 469.44, model IV = 467.93, model V = 472.58 for throat). However, given limitations in sample size and high variation within localities even in the extremes, cline parameter estimates for these traits should be taken with caution.

We noted high variation in underpart hue among individuals collected at the same site. In addition, in some

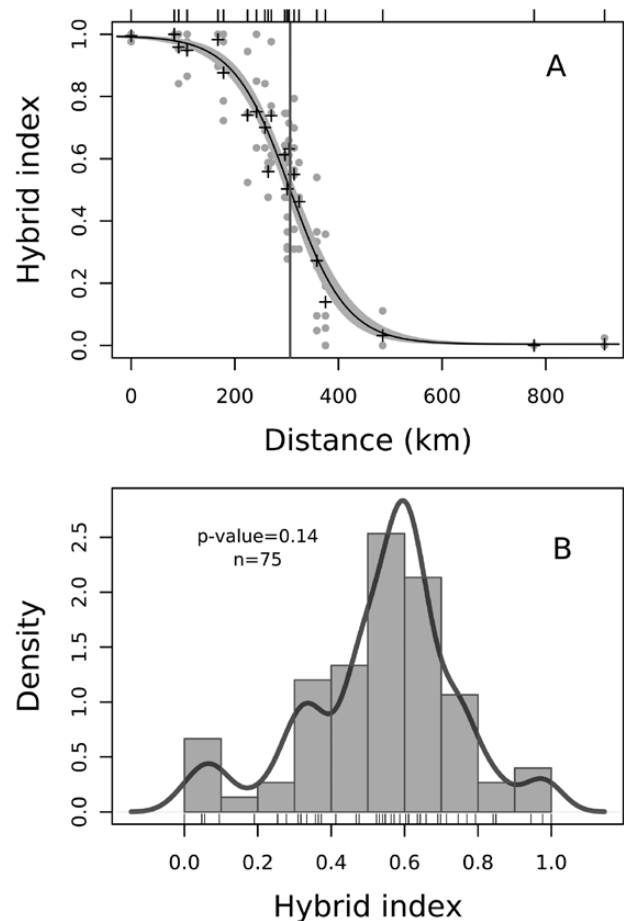


FIGURE 4. Variation of head coloration exhibits a clear sigmoidal pattern of variation along distance, with intermediates being the most common in the hybrid zone. (A) Plumage cline describing the *Myioborus* hybrid zone. The hybrid index was built taking into account only head plumage patches, where 1 (one) represents a typical *M. m. ruficoronatus* and 0 (zero) a typical *M. o. chrysops*. Gray dots represent individual score values, crosses are mean values per locality and the vertical line is the estimated center. (B) Unimodal distribution of plumage phenotypes as measured by the hybrid index (localities 8–19).

cases, the mean hue varied markedly between neighboring localities within the hybrid zone. For example, specimens from Puerres (arrow, Figure 6B) exhibited the highest values of mean hue and differed considerably from those of nearby localities. This could reflect that alongside potential clinal variation in chest hue, underpart coloration also varies patchily in space. Because the underparts showed a reflectance peak in UV wavelength (Appendix Figure 7), variation in hue as measured with reflectance spectrometry is not readily perceived by the human eye. However, there was also subtle observable variation in underpart coloration. For example, all individuals at Romerillo (Figure 6E, locality 14 in Figure 1) had underparts with a more orange tinge than most specimens from elsewhere. We also

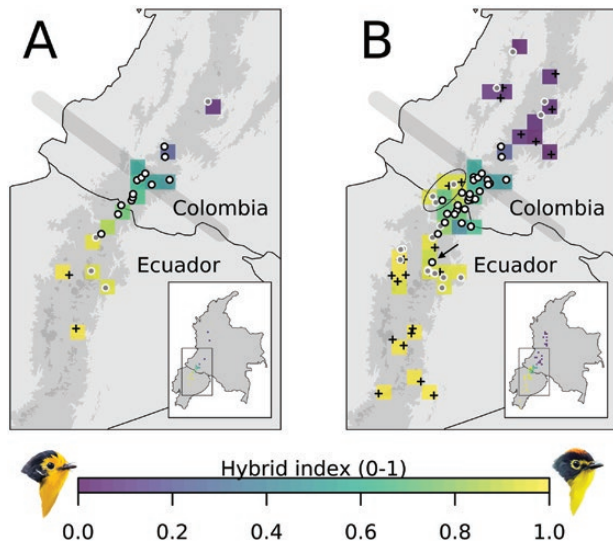


FIGURE 5. Distribution of individuals showing intermediate phenotypes (i.e. hybrids) (A) based on specimens included in cline analyses and (B) adding other specimens collected in the area since 1897. Intermediates are concentrated in extreme southern Colombia and northernmost Ecuador. The colored squares indicate the mean hybrid indices per $0.25^\circ \times 0.25^\circ$ pixel (0 = *M. o. chrysops*, 1 = *M. m. ruficoronatus*). Crosses correspond to localities where all specimens were confidently assigned to either *M. o. chrysops* or *M. m. ruficoronatus*, hence with zero intermediates. White circles correspond to localities with at least 1 hybrid, whereas gray circles to localities where at least 1 specimen was not confidently classified as hybrid or either taxon. The approximate location of the hybrid zone center inferred by cline analyses (Figure 4) is indicated by a transversal gray bar. The ellipse corresponds to western reaches of the Andes in the Colombia-Ecuador border, and the arrow to Oyacachi.

found several specimens from the southernmost localities (Parroquia El Triunfo, locality 1 in Figure 1; Paschoa, locality 2 in Figure 1) exhibiting a marked contrast between a yellow abdomen with an almost orange chest and throat which was lacking in other areas (Figure 6E).

The extent of white in the tail did not clearly vary along the sampling transect (Figure 6A). Although we noticed considerable variation among individuals, with some specimens having the third outer rectrix (R4) almost entirely white and others having it entirely black, no differences existed between the extremes of the transect (Figure 6D). Furthermore, we did not find any clear pattern with distance along the sampling transect with respect to the amount of white in each feather (Appendix Figure 10).

DISCUSSION

The understanding of the origin of Andean bird diversity has been mostly focused on assessing the role of

geographic and ecological barriers promoting allopatric differentiation, whereas studies of other processes such as those that occur when formerly isolated populations come into contact (e.g., hybridization and introgression; Dubay and Witt 2014, Morales-Rozo et al. 2017, Winger 2017) have been notably scarce. In addition to describing how genetic structure and geographic variation in plumage in the *M. ornatus*–*M. melanocephalus* complex is related to topographical barriers along the Andes, here we explored processes that may act upon secondary contact describing a hybrid zone between *M. o. chrysops* and *M. m. melanocephalus* using mtDNA and plumage data. Genetic structure in the *M. ornatus*–*M. melanocephalus* complex partially coincided with one physical barrier (i.e. the Marañón River Valley). However, geographic variation in plumage was not always associated with genetic structure in mtDNA. Along a continuous cloud forest belt in the area where the ranges of *M. o. chrysops* and *M. m. ruficoronatus* abut, we found a 200 km cline in head coloration. The existence of a broad hybrid zone where individuals with intermediate plumage are abundant suggests that hybridization between these taxa has been extensive.

Phylogeography and Population Structure

Our comprehensive geographic sampling revealed that mtDNA genetic structure in the *M. ornatus*–*M. melanocephalus* complex does not clearly coincide with current taxonomy. Specifically, we found that haplotypes from *M. m. ruficoronatus* specimens from Ecuador appear to be more closely related to haplotypes from both subspecies of *M. ornatus* than to those of southern conspecific populations, a result consistent with previous work with more limited sampling (Pérez-Emán 2005). Genetic structure was shallow, and only clearly associated with 1 topographic discontinuity of the cloud forest belt. We found evidence of mtDNA differentiation across the Marañón River Valley, a dry area that dissects the cloud forest distribution of *M. melanocephalus* and coincides with the transition between rufous-crowned (*M. m. griseonuchus*) and black-crowned (*M. m. malaris*) forms (Zimmer 1949, Curson et al. 1994). This arid valley is important for differentiation in many other cloud forest birds (Bates and Zink 1994, Chaves et al. 2011, Gutiérrez-Pinto et al. 2012, Winger and Bates 2015), and likely acts as a strong barrier for *Myioborus* taxa restricted to humid high-elevation forests and scrub (Curson et al. 1994). The exception are 3 specimens from south of the Marañón (Amazonas, Peru), corresponding to the black-crowned subspecies *M. m. malaris*, which clustered with individuals of *M. m. griseonuchus* and *M. m. ruficoronatus* occurring north of the Marañón Valley. This pattern might reflect trans-Marañón introgression (Winger 2017) or incomplete lineage sorting (Maddison and Knowles 2006). Other

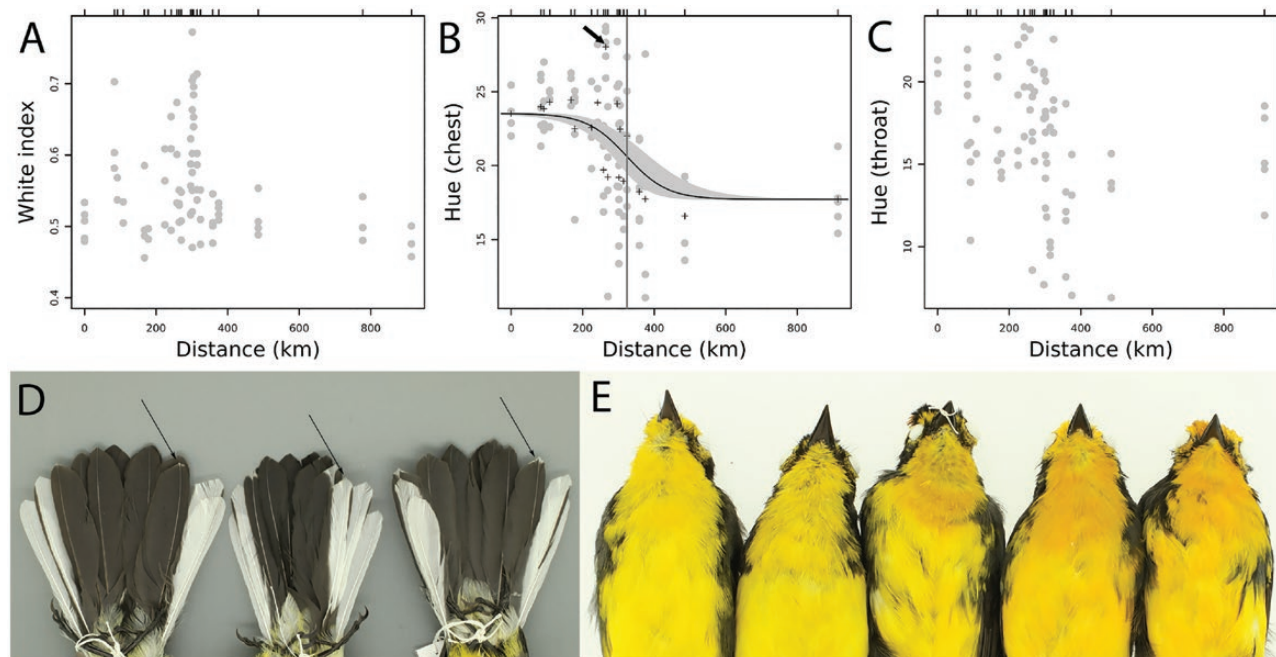


FIGURE 6. Body plumage coloration in *M. o. chrysops*–*M. m. ruficoronatus* does not show clear geographic variation, except for chest hue. A fitted cline is shown only for chest hue, the only trait seemingly following a sigmoidal pattern of variation. (A) Variation in the amount of white in tail feathers (white index) across the sampling transect. (B, C) Variation in hue of the chest and throat. In panel (B), crosses represent means by locality and the arrow indicates the highest locality mean. (D) Examples of specimens to illustrate the variation in the amount of white in the tail, with arrows pointing to the third outer rectrix (R4) indicating that is almost completely white in the specimen in the middle (from Vereda Romerillo, locality 14) and almost entirely black in specimens from the extremes of the transect (La Montañita, locality 22; El Triunfo, locality 1). (E) Specimens showing visible variation in underpart coloration. The two in the left (from San José de Córdoba, locality 13; Yanayacu Biological Station, locality 3) are bright yellow, whereas the two on the right (from Romerillo, locality 14) show an orange tinge. The specimen in the middle (from El Triunfo, locality 1) has an orange hue in the throat and chest, contrasting with a bright yellow abdomen.

geographic features potentially influencing genetic structure are dry valleys in southern Ecuador, such as the Upano and Zamora River valleys (Krabbe 2008, Bonaccorso 2009), which may account for the differentiation between most *M. m. ruficoronatus* and *M. m. griseonuchus*. However, additional sampling in southern Ecuador is required to better understand the complex patterns of genetic and head color variation in the area (Zimmer 1949).

In contrast to patterns of moderate genetic structure associated with a physical and ecological barrier in the south (i.e. divergence across the Marañón in northern Peru), we found a very low genetic structure in mtDNA in the *M. ornatus*–*M. melanocephalus* complex from central Ecuador to Venezuela, a region where a wide variety of plumage forms occur. For example, individuals of the phenotypically distinct, white-cheeked *M. o. ornatus* shared haplotypes with “pure” *M. o. chrysops*, the northern subspecies of *M. melanocephalus*, and individuals of the hybrid zone. We did not find any evidence of considerable genetic structure across the hybrid zone, a pattern contrasting with numerous avian hybrid zones characterized by sharp clines of haplotype

or allele frequencies in mtDNA markers (Irwin et al. 2009a, Carling and Zuckerberg 2011, Miller et al. 2014; but see Morales-Rozo et al. 2017). The lack of ND2 divergence may reflect rapid and recent diversification in the *M. ornatus*–*M. melanocephalus* complex in the northern Andes resulting in incomplete lineage sorting (Maddison and Knowles 2006). Haplotype sharing among all these northern taxa suggests that differences in coloration patterns may have evolved rapidly (Ödeen and Bjorklund 2003, Milá et al. 2007, Campagna et al. 2012, Harris et al. 2018, Palacios et al. 2019). An alternative explanation is mitochondrial introgression, which may lead to discordant patterns of variation in mtDNA relative to nuclear DNA and plumage (Weckstein et al. 2001, Irwin et al. 2009b, Brelsford et al. 2011). Genomic data are necessary to evaluate whether incomplete lineage sorting or mtDNA introgression account for the observed patterns, although the existence of numerous ND2 haplotypes is arguably evidence against the latter hypothesis. In addition, genomic-wide data would allow evaluating the relative effects of isolation-by-distance (Wright 1943) and physical barriers in shaping genetic structure along the

linear and discontinuous distribution of this complex (Harvey and Brumfield 2015, Kopuchian et al. 2020), a test not feasible with the present mtDNA dataset (Teske et al. 2018).

The Hybrid Zone

The geographic area where the *Myioborus* hybrid zone is centered (Nariño Department; Figure 1B, localities 14–16; Figure 5) has no clear topographic or climatic breaks potentially influencing the continuity of the cloud forest belt in the present (Graham et al. 2010). Diversification within the *M. ornatus*–*M. melanocephalus* complex likely occurred during the Pleistocene (Pérez-Emán 2005, Cuervo 2013), a period characterized by elevational shifts of vegetation belts (Haffer 1974, Hooghiemstra and van der Hammen 2004, Cárdenas et al. 2011) leading to changing patterns of fragmentation of cloud forest along the Andes (Ramírez-Barahona and Eguarte 2013). Therefore, the connectivity of cloud forest in this area may have become interrupted, promoting allopatric divergence (Graves 1985, 1988), and later reestablished allowing secondary contact between forms differentiated in plumage. Palynological evidence suggest that the area where the *Myioborus* hybrid zone is centered has experienced major changes in the connectivity of some vegetation belts (Flantua et al. 2014, 2019). Remarkably, this broad area coincides with a concentration of phylogeographic breaks in other cloud forest birds (e.g., *Myiothlypis coronata*, *Hellmayrea gularis*, *Synallaxis unirufa*; Cuervo 2013, Prieto-Torres et al. 2018). However, the divergence of genetic groups across the area appears to have happened at different moments. This suggests that cloud forest connectivity could have been lost multiple times, promoting divergence across multiple Andean taxa, a hypothesis to be tested in a comparative phylogeography framework (Moritz et al. 2009).

Although we imply that the *Myioborus* hybrid zone formed by secondary contact, an alternative explanation for the observed cline in head coloration is primary intergradation (Endler 1977). These alternative scenarios are difficult to distinguish based on current patterns of variation (Price 2008, Harrison and Larson 2016). We argue that the most plausible scenario is secondary contact because primary intergradation is expected to occur along environmental gradients with contrasting selective pressures in the extremes, which are not clearly present along the cloud forest band that *M. ornatus* and *M. melanocephalus* inhabit (except perhaps for subtle variation in temperature seasonality; Graves 1991), and the hybrid zone does not coincide with an abrupt environmental transition (Graham et al. 2010).

Regardless of the origin of the hybrid zone, an emerging question is whether some form of selection maintains the integrity of *M. o. chrysops* and *M. m. ruficoronatus*. With

free interbreeding, the *Myioborus* hybrid zone might represent a transient stage leading to the homogenization of these plumage forms. In contrast, with selection against hybrids balanced by dispersal of parents into the zone, the hybrid zone should remain stable and relatively narrow (i.e. a tension zone; Barton and Hewitt 1985). Selection might further lead to the completion of reproductive isolation via reinforcement (Liou and Price 1994, Ortiz-Barrientos et al. 2009). Our field observations and examination of specimens revealed that pure plumage forms (*M. o. chrysops* and *M. m. ruficoronatus*) do not overlap and that all individuals are intermediate in the center of the hybrid zone. This pattern, together with the high diversity of intermediate plumage phenotypes featuring various combinations of facial traits typical of pure *M. o. chrysops* and *M. m. ruficoronatus*, likely reflects that many individuals are advanced-generation hybrids or product of backcrosses (Price 2008), a pattern contrasting with hybrid zones in which selection acts strongly against hybrids (Pulido-Santacruz et al. 2018). Thus, our data suggests that hybrids are viable and fertile, as expected when divergence time between hybridizing taxa is low (Price and Bouvier 2002). Furthermore, we found that the *Myioborus* hybrid zone is unimodal (parental forms are the least common near the center), suggesting extensive hybridization and little disruptive selection (Jiggins and Mallet 2000, Gay et al. 2008). However, even unimodal hybrid zones could be characterized by selection against hybrids (Szymura and Barton 1991, Jiggins and Mallet 2000) or by assortative mating acting in the borders (Smadja et al. 2004), scenarios that could be at play in the *Myioborus* hybrid zone but cannot be evaluated based solely on patterns of plumage variation.

Although phenotypic evidence suggests that hybridization is ongoing and hybrids are viable, a more direct evaluation of selection against hybrids should compare our estimated cline width of ~200 km for head coloration with cline widths expected under neutral diffusion ($w = 2.51\sigma\sqrt{t}$, Barton and Gale 1993). Specifically, the neutral cline should be wider than observed clines if some form of selection against hybrids is acting (Toews et al. 2011, Baldassarre et al. 2014, Seneviratne et al. 2016). Although estimates of generation time (2 years; Pérez-Emán et al. 2010) and natal dispersal distances (935 m for females and 485 m for males; Mumme 2015) are available for another *Myioborus*, we lack accurate estimates of the timing since secondary contact, precluding an assessment of cline width under neutral diffusion. However, compared to other warbler hybrid zones with similar characteristics (i.e. excluding mosaic hybrid zones; Vallender et al. 2007) the *Myioborus* hybrid zone is wide (*Geothlypis tolmiei* × *G. philadelphia*: 130 km, Irwin et al. 2009a; *Setophaga townsendi* × *S. virens*: 40–87 km, Toews

et al. 2011; *Setophaga coronata auduboni* × *S. c. coronata*: 132 km, Brelsford and Irwin 2009; *Setophaga occidentalis* × *S. townsendi*: 100–125 km, Rohwer and Wood 1998, Rohwer et al. 2001). This difference in width between the *Myioborus* hybrid zone and other hybrid zones involving warblers appears especially remarkable because *Myioborus* are tropical resident birds expected to disperse over shorter distances than migratory species like all of the above (Paradis et al. 1998). In sum, the data suggest that either selection against hybrids is weaker or that the *Myioborus* hybrid zone is older compared to other warbler hybrid zones. If one assumes that patterns of variation are explained by neutral diffusion, and uses values of dispersal distance for females and generation times reported for closely related *Myioborus*, then this hybrid zone would be between 13,386 and 22,154 years old. Alternatively, assuming values of dispersal distances estimated for males, which are much lower, the hybrid zone would be between 49,751 and 82,337 years old.

In contrast to head color patterns, which showed a clear cline, we only found a rather clear (i.e. sigmoidal) cline in one of the 3 body color traits measured: chest coloration, as described by hue. Both head coloration and chest hue clines are concordant in having similar centers and widths. In addition, our data showed that the yellow underpart coloration can vary markedly among individuals within sites or among neighboring populations, which could be partially due to subtle differences in access to dietary carotenoids (Hill 1993, McGraw et al. 2001, Hill et al. 2002, Toews et al. 2017). Phenotypic extremes for underpart hue were exhibited by hybrid zone individuals and not by “pure” individuals at the transect extremes, which reflects a sampling bias given lower sample sizes in the extremes or, potentially, the expression of transgressive phenotypes in hybrids (Rieseberg et al. 1999), a hypothesis to be tested with a more balanced sampling design. Lastly, we found no clear differentiation across the sampling transect in the extent of white on tail feathers, a trait showing clear geographic variation with potential adaptive value in another *Myioborus* (Pérez-Emán et al. 2010). The clade of high-elevation *Myioborus*, which comprises 10 species including *M. ornatus* and *M. melanocephalus*, is characterized by an overall similarity in body coloration and striking variation in the head color pattern (Pérez-Emán 2005). Plumage patches in the head are often involved in intraspecific communication and mate choice (Andersson et al. 1998, Rémy et al. 2010), and exhibited during displays in many bird species (Marchetti 1992, Price and Pavelka 1996, Scott and Deag 1998) suggesting that such traits might be under strong social selection and are therefore likely to evolve rapidly (West-Eberhard 1983). Detailed documentation on mating and territorial displays in these *Myioborus* would allow examining how color patches in the head are shown

and evaluated in reproductive and social contexts. Also, differences in these color patches might be controlled by few genetic regions with large phenotypic effects (Uy et al. 2009, Poelstra et al. 2014, Toews et al. 2016, Campagna et al. 2017) facilitating rapid divergence when populations become isolated.

Implications for Taxonomy

Some authors (Meyer de Schauensee 1964, Robbins et al. 1994) have called attention to intermediate phenotypes in southern Colombia and northern Ecuador, which could suggest that *M. ornatus* and *M. melanocephalus* might be conspecific. In contrast, others used phenotypic scores to propose splitting the nominate, white-faced *M. ornatus* from the yellow-faced taxon *M. o. chrysops* (Handbook of the Birds of the World and BirdLife International 2020). Our findings of hybridization with phenotypic intergradation across an extensive region of the Andes, and a high frequency of intermediate phenotypes suggests that *M. o. chrysops* and *M. m. ruficoronatus* likely experience extensive gene flow and are not reproductively isolated. Hence, under the Biological species concept (Mayr 1942), these populations might arguably be considered members of a single species that diverged in allopatry and are now merging together upon secondary contact, with a difference in plumage being unimportant for species recognition. More broadly, mtDNA differentiation is low across this complex of 7 subspecies, potentially a product of recent differentiation. Forthcoming analyses of patterns of variation across the nuclear genome in this complex will put us in a better position to comprehensively assess whether taxonomic changes are deemed necessary. Such work should also involve sampling of topotypic specimens because the type locality of *M. m. ruficoronatus* lies within the known range of *M. o. chrysops* (Salvin 1837, Meyer de Schauensee 1964) and the type may be a hybrid.

Conclusions

Myioborus ornatus and *M. melanocephalus* are 2 Andean birds characterized by low genetic divergence in mtDNA and outstanding geographic variation in color patterns. Phenotypic evidence suggests that these species have hybridized extensively where their 2 extreme subspecies (*M. o. chrysops* and *M. m. ruficoronatus*) meet, forming a hybrid zone along ~200 km, where individuals with intermediate plumage are the most common. Future studies should focus on evaluating whether selection against hybrids may be maintaining the integrity of *M. m. ruficoronatus* and *M. o. chrysops*, 2 young taxa with striking differences in head color pattern. Also, as suggested by specimens collected prior to our study, a systematic sampling across the western side of the Andes in the Colombia-Ecuador border might reveal patterns of variation of head coloration to be

more complex than described herein. Collecting genome-wide data will allow us to explicitly evaluate the existence of selection against hybrids (e.g., by allowing for the calculation of linkage disequilibrium values; Brelsford and Irwin 2009, Taylor et al. 2012), describe the genetic composition of the hybrid zone (i.e. the proportion of individuals in different hybrid classes; Toews et al. 2018, Lipshutz et al. 2019) and identify genomic regions of limited introgression (Poelstra et al. 2014, Toews et al. 2016, Walsh et al. 2016, Baiz et al. 2019), potentially associated with incipient reproductive isolation. Our specimen-based description provides a starting point for additional research on the dynamics of this *Myioborus* hybrid zone, a valuable step for future assessments of how species are formed and maintained over time along the species-rich Andean mountains.

SUPPLEMENTARY MATERIAL

Supplementary material is available at *Ornithology* online.

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Author contributions: L.N.C., A.M.C., and C.D.C. conceived the idea. L.N.C., A.M.C., E.B., J.L.P. and C.D.C. designed the methodology. L.N.C., A.M.C., A.M., and C.D.C. conducted fieldwork. L.N.C. and M.C. conducted laboratory work and assembled DNA sequences. L.N.C. and A.M.C. collected data from museum specimens. L.N.C. analyzed the data. C.D.C., E.B., and C.W. contributed to substantial materials and resources. All authors contributed to write the manuscript, led by L.N.C., A.M.C., and C.D.C.

Data availability: Genetic data are available in GenBank (accession numbers: AY968913, AY968912, AY968905, AY968904, AY968900, AY968898, AY968899, AY968935, AY968933, AY968932, and MT996802 - MT997008). Plumage data are available from the author at Céspedes-Arias et al. (2021).

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APPENDIX TABLE 3. Sampling localities across the hybrid zone and putative allopatric populations (1, 21, and 22). Numbers correspond to those in Figure 1. Sampling sizes are shown for the number of specimens collected in this study, and specimens with plumage data included in cline analyses.

	Locality	Country	Department/ Province	Specimens (n)	Cline (n)	Distance in transect (km)	Latitude	Longitude
1	Parroquia El Triunfo	Ecuador	Tungurahua	5	5	0	-1.316	-78.403
2	Refugio de Vida Silvestre Pasocha	Ecuador	Pichincha	5	5	83	-0.425	-78.517
3	Yanayacu Biological Station	Ecuador	Napo	5	5	92	-0.640	-77.913
4	Termas de Papallacta	Ecuador	Napo	5	4	108	-0.358	-78.151
5	Hacienda Zuleta	Ecuador	Imbabura	6	6	168	0.199	-78.072
6	Bosque Nueva America	Ecuador	Imbabura	5	5	178	0.260	-77.982
7	La Sofia Road	Ecuador	Sucumbios	1	0	215	0.439	-77.587
8	Guandera Biological Station	Ecuador	Carchi	4	4	225	0.589	-77.704
9	Vereda El Mirador	Colombia	Nariño	5	5	242	0.736	-77.652
10	Vereda La Cumbre	Colombia	Nariño	5	4	258	0.815	-77.480
11	Vereda La Esperanza	Colombia	Nariño	5	5	264	0.867	-77.460
12	Corregimiento Chapal	Colombia	Nariño	6	5	270	0.923	-77.454
13	Vereda San José de Córdoba and Bosque El Zanjón	Colombia	Nariño	5	5	297	1.155	-77.389
14	Laguna de la Cocha	Colombia	Nariño	8	18	301	1.080	-77.145
15	Vereda San Felipe	Colombia	Nariño	5	5	305	1.198	-77.315
16	Alto de Daza	Colombia	Nariño	7	8	314	1.266	-77.253
17	Vereda La Siberia and Minchay	Colombia	Putumayo	5	5	324	1.161	-76.847
18	Vereda La Palma	Colombia	Nariño	6	6	358	1.543	-76.923
19	Cerro de la Campana and Vereda Ramal Alto	Colombia	Nariño	5	5	375	1.715	-76.934
20	Vereda Riosucio	Colombia	Cauca	5	5	485	2.464	-76.210
21	Villamaria	Colombia	Caldas	4	4	778	5.003	-75.421
22	Finca La Montañita	Colombia	Antioquia	6	6	913	6.456	-75.601
				113	120			

APPENDIX TABLE 4. Genetic differentiation between groups in the *M. ornatus*–*M. melanocephalus* complex. Numbers below the diagonal correspond to the F_{ST} values that are in bold if P -value < 0.05. Above the diagonal, mean Nei-genetic distances between defined groups. These genetic distances and F_{ST} values were calculated using the classification for *M. m. griseonuchus* and *M. m. ruficoronatus* based on our examination of specimens.

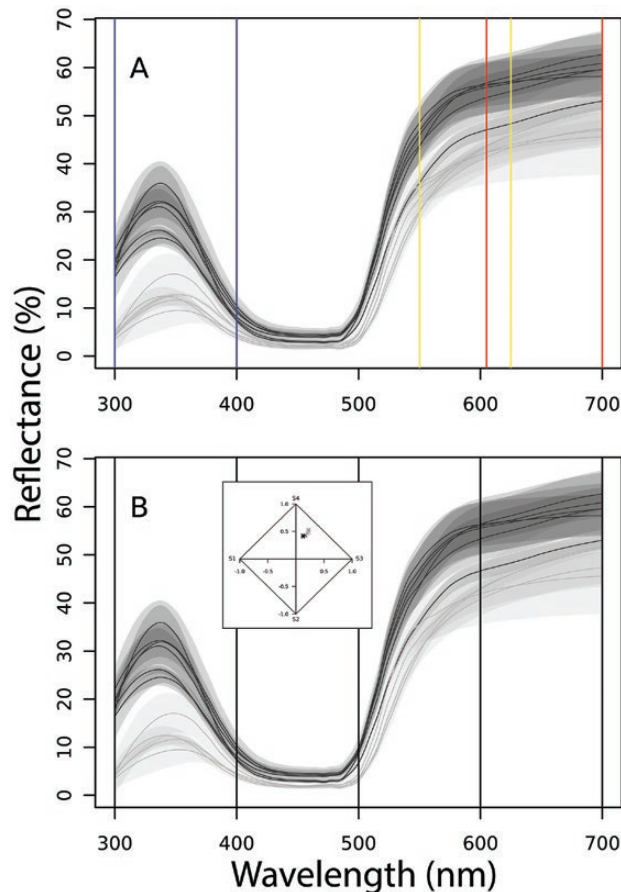
	<i>M. o. ornatus</i>	<i>M. o. chrysops</i>	Hybrid zone	<i>M. m. ruficoronatus</i>	<i>M. m. griseonuchus</i>	<i>M. m. malaris</i>	<i>M. m. melanocephalus</i>	<i>M. m. bolivianus</i>
<i>M. o. ornatus</i> ($n = 26$)	–							
<i>M. o. chrysops</i> ($n = 15$)	0.068	0.006	0.004	0.008	0.051	0.071	0.095	0.095
Hybrid zone ($n = 109$)	0.026	–	0.002	0.007	0.053	0.072	0.096	0.097
<i>M. m. ruficoronatus</i>	0.066	0.009	–	0.004	0.047	0.069	0.094	0.095
($n = 11$)		0.069	0.015	–	0.033	0.062	0.092	0.092
<i>M. m. griseonuchus</i>	0.271	0.294	0.149	0.150	–	0.076	0.111	0.109
($n = 14$)								
<i>M. m. malaris</i> ($n = 4$)	0.256	0.365	0.085	0.231	0.203	–	0.016	0.034
<i>M. m. melanocephalus</i>	0.418	0.504	0.191	0.375	0.353	0.086	–	0.026
($n = 8$)								
<i>M. m. bolivianus</i>	0.424	0.489	0.220	0.368	0.355	0.149	0.152	–
($n = 10$)								

APPENDIX TABLE 5. Population genetic statistics for subspecies and the hybrid zone geographic group. Values were calculated using both classification schemes for birds in southern Ecuador. Grouping A corresponds to the classification based on historical reports and grouping B to the classification based on our own assessment of specimens. We provide two different *P*-values for Tajima's *D*: (1) "normal" which is calculated assuming a normal distribution for *D*; and (2) "beta" which is calculated assuming a beta distribution for *D*.

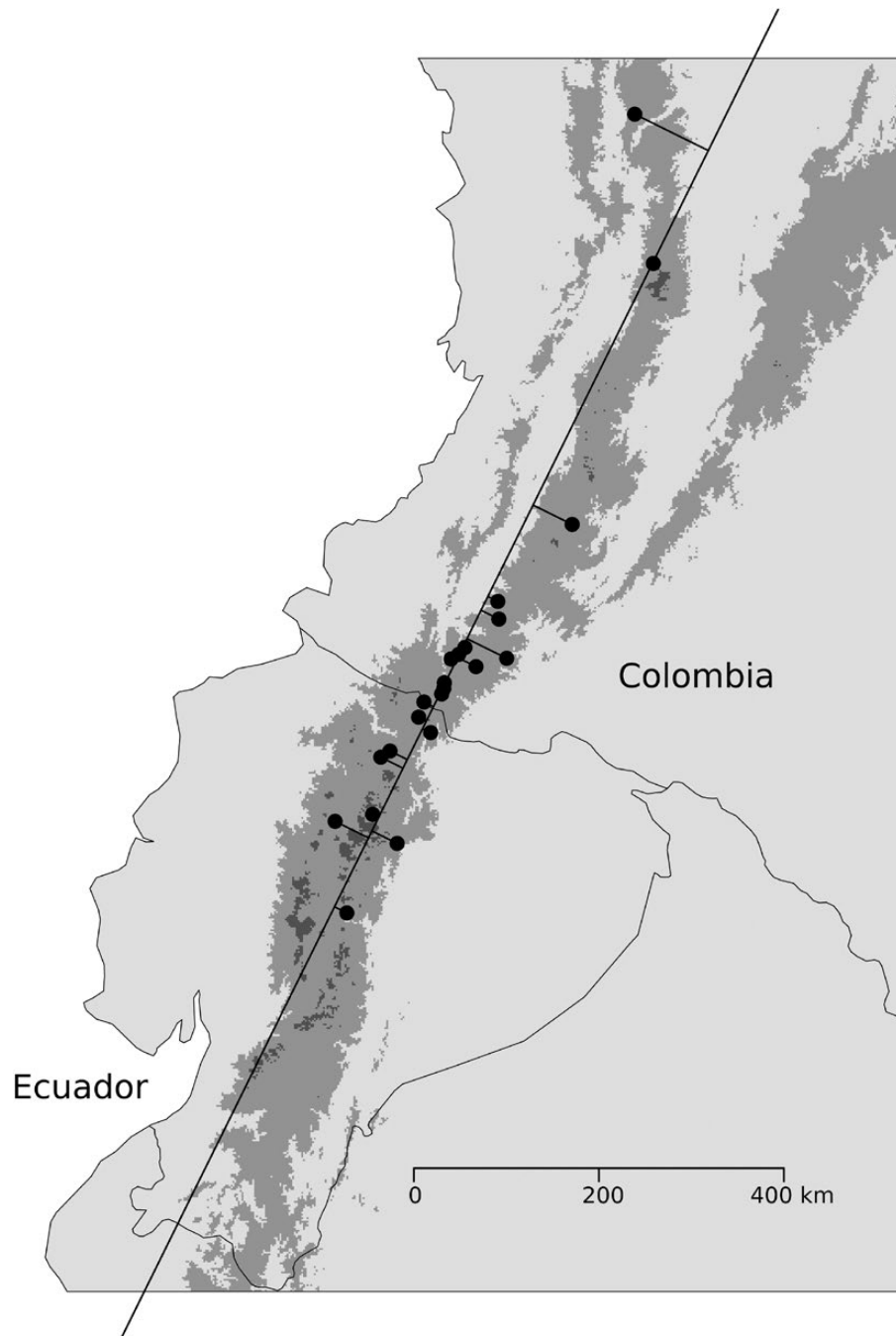
	Nucleotide diversity (π)	Haplotype diversity	Tajima's <i>D</i>	<i>P</i> -value (normal)	<i>P</i> -value (beta)
Grouping (A)					
<i>M. o. ornatus</i> (<i>n</i> = 26)	0.002	0.846	−1.433	0.152	0.146
<i>M. o. chrysops</i> (<i>n</i> = 15)	0.002	0.848	−1.040	0.298	0.327
Hybrid zone (<i>n</i> = 109)	0.002	0.811	−2.020	0.043	0.019
<i>M. m. ruficoronatus</i> (<i>n</i> = 16)	0.006	0.983	−1.083	0.279	0.303
<i>M. m. griseonuchus</i> (<i>n</i> = 9)	0.002	0.750	−0.035	0.972	0.991
<i>M. m. malaris</i> (<i>n</i> = 4)	0.007	1.000	−0.845	0.398	0.766
<i>M. m. melanocephalus</i> (<i>n</i> = 8)	0.005	0.964	−1.337	0.181	0.188
<i>M. m. bolivianus</i> (<i>n</i> = 10)	0.005	0.978	−0.798	0.425	0.484
Grouping (B)					
<i>M. o. ornatus</i> (<i>n</i> = 26)	0.002	0.846	−1.433	0.152	0.146
<i>M. o. chrysops</i> (<i>n</i> = 15)	0.002	0.848	−1.040	0.298	0.327
Hybrid zone (<i>n</i> = 109)	0.002	0.811	−2.020	0.043	0.019
<i>M. m. ruficoronatus</i> (<i>n</i> = 11)	0.004	0.982	−1.402	0.161	0.162
<i>M. m. griseonuchus</i> (<i>n</i> = 14)	0.005	0.890	0.208	0.836	0.827
<i>M. m. malaris</i> (<i>n</i> = 4)	0.007	1.000	−0.845	0.398	0.766
<i>M. m. melanocephalus</i> (<i>n</i> = 8)	0.005	0.964	−1.337	0.181	0.188
<i>M. m. bolivianus</i> (<i>n</i> = 10)	0.005	0.978	−0.798	0.425	0.484

APPENDIX TABLE 6. Correlation between colorimetric variables (Pearson's *r*) describing underpart coloration. Values above of the diagonal correspond to the throat and below to chest.

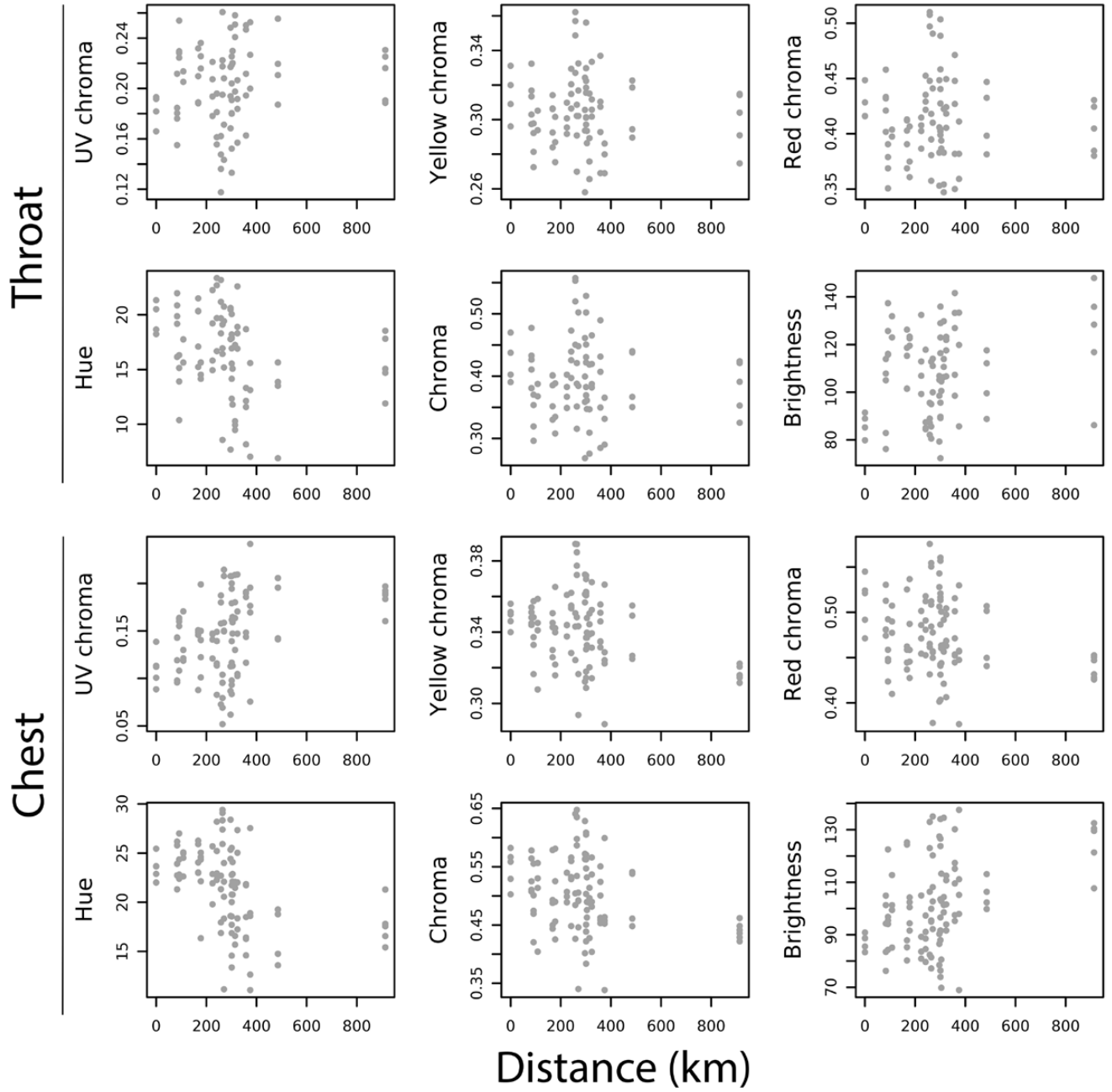
	S1U	S1Y	S1R	H	C	B
UV chroma (S1U)	–	−0.907	−0.936	−0.880	−0.919	0.431
Yellow chroma (S1Y)	−0.884	–	0.955	0.741	0.985	−0.406
Red chroma (S1R)	−0.913	0.925	–	0.706	0.978	−0.433
Hue (H)	−0.801	0.586	0.523	–	0.734	−0.421
Chroma (C)	−0.939	0.959	0.979	0.613	–	−0.443
Brightness (B)	0.653	−0.557	−0.583	−0.547	−0.599	–



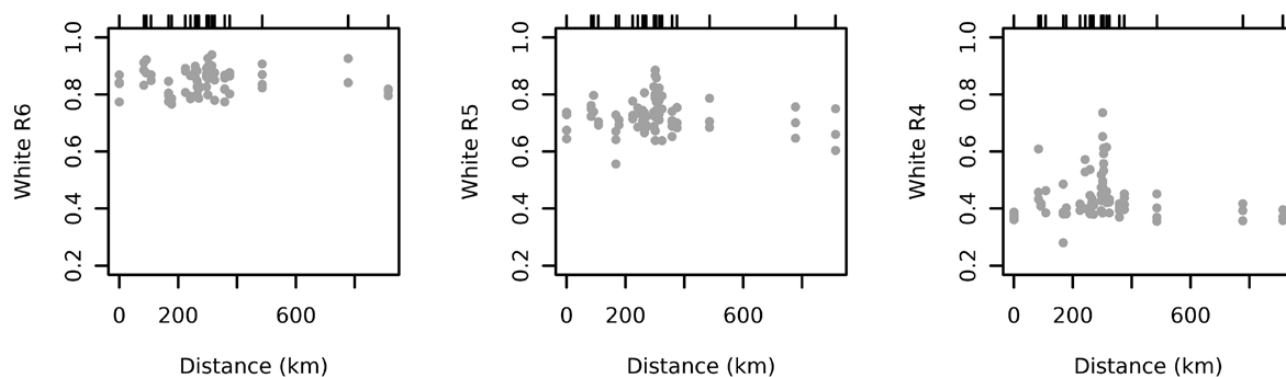
APPENDIX FIGURE 7. Example of reflectance spectra corresponding to specimens from populations at the extremes of the sampling transect. Each line corresponds to the average of reflectance spectra taken from the chest of a specimen and the shading to its standard deviation. Dark gray corresponds to specimens collected in the northernmost locality (La Montaña, locality 22), and light gray to specimens collected in the southernmost locality (El Triunfo, locality 1). The same reflectance spectra are shown in both panels, but they differ in markings related to how colorimetric variables were calculated. **(A)** Vertical lines show the ranges used to calculate the following colorimetric variables: UV chroma (S1U, purple lines), yellow chroma (S1Y, yellow lines), and red chroma (S1R, red lines). **(B)** Vertical lines denote the 100 nm divisions used to locate each averaged spectrum in the segment classification color space. Crosses in the inset show the projection of these reflectance spectra on the axes of segment classification.



APPENDIX FIGURE 8. Localities of collection of specimens (black dots) across the hybrid zone and allopatric populations, depicting the regression line used to calculate the distance for geographic cline analyses. The black line that goes parallel to the Andes corresponds to the regression line (where x = longitude and y = longitude). Distances were calculated from the southernmost locality to the intersect of the short perpendicular lines that start at each sampling locality.



APPENDIX FIGURE 9. Variation of 6 variables describing underpart coloration across our sampling transect for both throat and chest. Panels for hue of throat and chest are shown in [Figure 6](#).



APPENDIX FIGURE 10. Variation across our sampling transect in the extent of white in each of the outer tail feathers examined. The amount of white per feather was calculated by averaging the extent of white on both vanes and dividing the result by the total vane length.